

SyncFeel: Wearable Multi-Modal Rhythm Stimulation for Attention Enhancement

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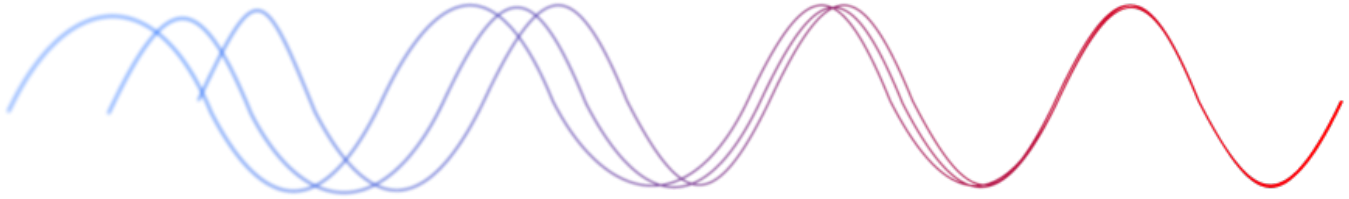


Figure 1: Visual allegory of brainwave entrainment as assisted by DataFeel Multi-Energy Technologies

Abstract

SyncFeel is a wearable platform for exploring multi-modal rhythmic stimulation for attention modulation. It consists of a lightweight, two-Dot bracelet that delivers synchronized vibration, light, and thermal pulses to subtly guide the wearer's cortical rhythms toward either heightened focus or calm reset. Our primary contributions are (1) a fully untethered implementation with DataFeel Dots, (2) the introduction of a novel, rapidly-responding thermal channel to the design space of attention wearables, and (3) a companion application that allows users to personalize the rhythmic patterns. This transforms the device into a platform for user-driven exploration. We demonstrate the system's potential with two research-informed presets designed to support focused work sessions and restorative attention breaks.

Keywords

Human-centered computing, interaction devices, haptic devices

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1 Introduction

Maintaining sustained attention is a critical challenge for knowledge workers, students, and creatives, as well as those actively seeking to improve their capacity for accessing and maintaining attentive states. A growing body of research suggests that rhythmic

light, sound, and haptic stimuli can improve cognition through mechanisms such as neural entrainment and by modulating autonomic nervous system function [8, 9]. Our aim is to create a fully untethered and personalizable platform that allows users to explore multi-modal rhythmic stimuli for attention modulation. We leverage the DataFeel Dots to introduce a novel thermal channel alongside traditional vibration and light, all of which can be customized in real-time by the user through a companion application.

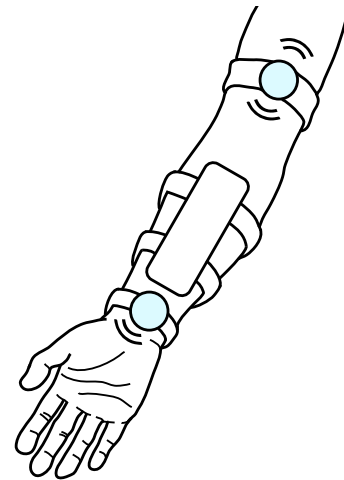


Figure 2: Concept art of SyncFeel

2 Motivation

Modern learners and knowledge workers fight a constant battle against distraction. While some software tools attempt to shield users from things like mobile app notifications, our ability to sustain deep, goal-directed attention ultimately rests on biological foundations. Brainwave entrainment refers to the process by which an

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external rhythmic stimulus causes neural oscillations to synchronize in frequency and phase with that stimulus, influencing perception and cognition. When a periodic external stimulus falls within the natural electroencephalogram (EEG) range (approx. 1–30 Hz), neuronal populations can phase-lock to that stimulus; this is the process of *brainwave entrainment* [6].

Recent work has demonstrated that brief wrist-worn vibrotactile burst trains at beta frequencies measurably improve task accuracy [8]. Although promising, this focused primarily on vibrotactile and auditory stimuli, leaving the opportunity to explore the role of the thermal modality.

Studies have validated that users can perceive spatial thermal cues on the wrist with high accuracy, especially for cold stimuli. This makes cold a prime candidate for a salient, non-distracting alerting cue [5]. Conversely, gentle warming is associated with relaxation and can be used to guide restorative practices like controlled breathing [7]. With the DataFeel Dev Kit, we see a chance to translate these lab insights into a fully wearable, field-ready attention coach. We aim to incorporate cooling thermal cues as both attention-cuing mechanisms to counter fatigue, as well as use the heating thermal cues for relaxation aid. SyncFeel explores these emerging areas of research to help users discover and create personalized patterns to enter and maintain states of focused flow.

2.1 Concept Overview and Use Cases

We envision SyncFeel as a minimalist two-Dot bracelet driven by one El Jefe controller, paired with a lightweight companion application. The system is demonstrated with two research-informed presets that serve as starting points for user exploration. The system can be adapted in the future to be used as an online closed-loop system using bio-signals, such as consumer EEG band and eye tracking devices if available.

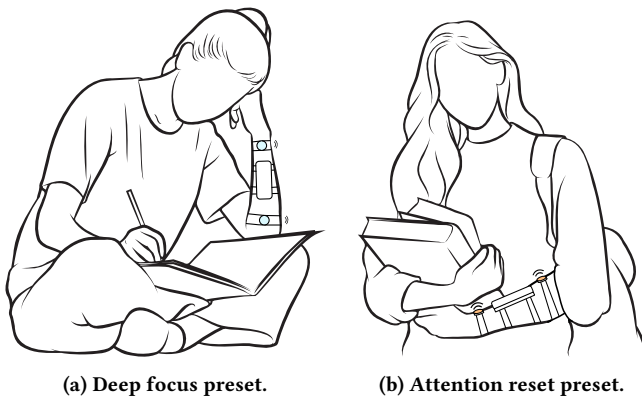


Figure 3: Proposed presets to induce deep focus with beta wave entrainment (a) and attention reset with alpha wave entrainment (b)

2.2 Deep-Work with Beta Wave Entrainment

Beta wave brain activity is associated with active concentration [4]. Entrainment aims to present a fast rhythmic cue so that cortical

networks synchronize to that frequency, boosting vigilance and reaction speed.

We propose a preset mode where 20 Hz vibration burst trains every 6 seconds, paired with a cool-white (6500K) LED micro-flicker. This high color temperature is chosen based on findings that it significantly improves task accuracy and reduces errors [1]. A 1°C cooling micro-pulse is also delivered every five minutes to counter vigilance drops.

2.2.1 Use Case. During a 25-minute Pomodoro [2], the wearer activates Beta-Focus. The Beta-Focus preset is designed to support sustained attention. It delivers sub-second haptic bursts intended to prime cortical beta oscillations, an unobtrusive LED flicker to reinforce the rhythm, and periodic cool spikes aimed at refreshing arousal. Unlike coffee or background music, the stimulus is silent and private.

2.3 Attention Reset with Alpha Wave Entrainment

Alpha wave brain activity is associated with relaxed alertness and quiet wakefulness [4]. We propose a preset mode with 8 Hz vibro taps coupled with warm-cool breathing waves (warm on inhale, cool on exhale) to enhance relaxation and synchronization, as well as soft, warm-amber (approximately 3300K) LED fades. This lower color temperature is linked to increased alpha wave activity and subjective visual comfort, promoting a state of relaxed alertness [3]. These cues guide a three-minute restorative pause before the next cognitive effort.

2.3.1 Use Case. Once task disruption has occurred, or task switching is necessary, it can be difficult to return to a focused state. For instance, teachers struggle to re-center restless students between activities. With the alpha entrainment preset mode, a three-minute synchronized warm-cool breathing guide and 8 Hz tactile taps can help students transition to a calmer, more focused state. The group shares a common rhythm without adding audio distraction.

3 User Personalization and Future Work

We also contribute a companion application that allows users to modify example patterns or create entirely new ones from scratch. We believe this is critical, as sensory preferences and effective stimuli can vary significantly between individuals.

User-driven customization is the foundation for our future work. While our current prototype focuses on an open-loop system, the architecture is designed to support a closed-loop future. By integrating real-time bio-signals from a consumer EEG headband or eye-tracker, the system could automatically calibrate patterns for an individual or perform online adaptation—adjusting the stimuli in real-time to help a user more effectively maintain a desired cognitive state.

4 Technical Approach

4.1 Hardware

- 2 DataFeel Dots per user: one on the inner wrist (peak vibro sensitivity) and one just below the elbow (thermal comfort,

outward-facing LEDs), for instance. This may be subject to change depending on feedback during development.

- El Jefe controller clipped to forearm strap. Bluetooth-MIDI link to host device; swappable 18650 cells provide approx. 2 hour runtime at 30 percent duty cycle.
- Wrist straps and 3D-printed arm shells from DataFeel STL library.

4.2 Software

A companion software application will serve as the rhythm engine. This application will feature a graphical user interface where users can select and customize patterns. The software will be responsible for generating the precise timing for the multi-modal stimuli and communicating these commands to the El Jefe controller via a wireless protocol.

5 Conclusion

SyncFeel makes two key contributions to the field of attention-shaping wearables. First, it introduces a novel and potentially powerful thermal channel, packing vibration, light, and rapid heating/cooling into a single, untethered device powered by DataFeel Dots. Second, it moves beyond fixed presets by providing a user-facing application for personalizing these multi-modal patterns. This reframes the system as a platform for user-driven exploration and paves the way for future closed-loop, adaptive interfaces that can learn and respond to an individual's unique biology.

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